moved slightly with respect to more distant background objects. The closer it is, the bigger the shift. Unfortunately, for anything outside our own galaxy, these apparent motions are too small to be perceived, and we need another method—a way to determine the distance of bright objects just from the properties of their light.

The key to everything from here on out is the concept of a standard candle, which we discussed briefly in the previous chapter. This is a kind of object (such as a star) that has some physical attribute that tells you its brightness. Then, by seeing how bright it *looks*, you can tell how far away it is. Kind of like having a light bulb with "60 Watts" written on it. You know how bright it should be, but you'll get less light from it when it's far away.

Of course, nothing in space has its brightness helpfully stamped on it. But we have something almost as good. The breakthrough discovery that first allowed us to use standard candles in astronomy was due to the astronomer Henrietta Swan Leavitt in the early 1900s.[fn7] Working at Harvard Observatory, she discovered that a certain kind of star known as a "Cepheid variable" brightened and dimmed in a predictable way. A Cepheid that's intrinsically brighter does slow, gradual pulsations, getting a little bit brighter and a little bit dimmer over a long period. A Cepheid that's intrinsically dimmer pulsates more quickly, with wide swings between its brightest and dimmest states.[fn8]

This discovery was revolutionary, and perhaps one of the most important in the history of astronomy, in that it let us finally measure the scale of the universe around us. It meant that anywhere a Cepheid could be seen, we could get a reliable distance and start to make a usable map. By measuring how quickly a Cepheid pulsed, and how bright it looked from here, Leavitt could tell you with great precision how bright it really was, and thus how distant.

How far does this get us? We can see Cepheid variable stars throughout the Milky Way and in nearby galaxies, so we can use parallax for the nearby ones, carefully calibrate the pulsation relationship, and then use the more distant ones to tell us the distances to other galaxies.

The next step in the distance ladder is a crucial one, but it's also where things get really messy, in every sense of the term. In the previous chapter, we mentioned that a certain kind of supernova can be used to measure distances. This kind of explosion, a Type Ia supernova, is what happens when a white dwarf star somehow picks up some mass from another,